

# AERO70015/96012 Mathematics 3 Problem Set #3

## Problem Sheet 3

### 3.1 PR 3.1

(a) Giving in each case the first three non-zero terms, and omitting "...", expand the function

$$f(z) = \frac{z}{(z-1)(2-z)}$$

in a Laurent series valid for:

- (i)  $0 < |z-1| < 1$  (ascending powers)
- (ii)  $|z-1| > 1$  (descending powers)
- (iii)  $0 < |z-2| < 1$  (ascending powers)
- (iv)  $|z-2| > 1$  (descending powers)

(b) Hence or otherwise determine the residues of  $f(z)$  at each of its poles.

(a) (i)  $z_0 = 1$

$$\begin{aligned} f(z) &= \frac{1}{z-1} + \frac{2}{2-z} = \frac{1}{z-1} + \frac{2}{1-(z-1)} \\ &= \frac{1}{z-1} + 2[1-(z-1)]^{-1} \\ &= \frac{1}{z-1} + 2[1 + (z-1) + (z-1)^2 + (z-1)^3 + \dots] \\ &= \frac{1}{z-1} + 2 \sum_{i=0}^{\infty} (z-1)^i \end{aligned}$$

(ii)  $z_0 = 1$

$$\begin{aligned} f(z) &= \frac{1}{z-1} + \frac{2}{2-z} = \frac{1}{z-1} - \frac{2}{(z-1)-1} = \frac{1}{z-1} - \frac{2}{z-1} \frac{1}{1-\frac{1}{z-1}} \quad \text{where } 0 < \left| \frac{1}{z-1} \right| < 1 \\ &= \frac{1}{z-1} - \frac{2}{z-1} [1 - (z-1)^{-1}]^{-1} \\ &= \frac{1}{z-1} - \frac{2}{z-1} [1 + (z-1)^{-1} + (z-1)^{-2} + \dots] \\ &= -\frac{1}{z-1} - \sum_{i=2}^{\infty} (z-1)^{-i} \end{aligned}$$

(iii)  $z_0 = 2$

$$f(z) = \frac{1}{z-1} - \frac{2}{z-2} = -\frac{2}{z-2} + \frac{1}{1+(z-2)}$$

$$\begin{aligned}
 \therefore f(z) &= -\frac{2}{z-2} + [1 + (z-2)]^{-1} \\
 &= -\frac{2}{z-2} + [1 - (z-2) + (z-2)^2 - (z-2)^3 + \dots] \\
 &= -\frac{2}{z-2} + \sum_{i=0}^{\infty} (-1)^i (z-2)^i
 \end{aligned}$$

(iv)  $z_0 = 2$

$$\begin{aligned}
 f(z) &= \frac{1}{z-1} - \frac{2}{z-2} = -\frac{2}{z-2} + \frac{1}{(z-2)+1} = -\frac{2}{z-2} + \frac{1}{z-2} \frac{1}{1+(z-2)^{-1}} \\
 &= -\frac{1}{z-2} + [- (z-2)^{-2} + (z-2)^{-3} + \dots] = -\frac{1}{z-2} + \sum_{i=2}^{\infty} (-1)^{i-1} (z-2)^{-i}
 \end{aligned}$$

(b)  $\text{Res}_{z=1} = 1$  and  $\text{Res}_{z=2} = -2$  for isolated pole at  $z=1$  and  $z=2$

### 3.2 PR 3.2

Expand the following functions in Laurent series valid in the regions given:

(a)  $\frac{e^z}{z(z^2+1)}$  for  $0 < |z| < 1$  (ascending powers, up to  $z^2$  term).

(b)  $\frac{1}{(z^2+1)(z^2+2)}$  for:

- (i)  $|z| > \sqrt{2}$  (descending powers, down to  $z^{-6}$  term)
- (ii)  $1 < |z| < \sqrt{2}$  (terms between  $z^{-6}$  and  $z^2$  inclusive)
- (iii)  $|z| < 1$  (ascending powers, up to  $z^4$ )

(a)  $z_0 = 0$

$$\begin{aligned}
 \frac{e^z}{z(z^2+1)} &= \frac{1+z+\frac{1}{2}z^2+\frac{1}{6}z^3+\frac{1}{24}z^4+\dots}{z(z^2+1)} \\
 &= \left(\frac{1}{z} + 1 + \frac{1}{2}z + \frac{1}{6}z^2 + \frac{1}{24}z^3 + \dots\right)(1+z^2)^{-1} \\
 &= \left(\frac{1}{z} + 1 + \frac{1}{2}z + \frac{1}{6}z^2 + \frac{1}{24}z^3 + \dots\right)(1-z^2+z^4-z^6+\dots) \\
 &= \frac{1}{z} + 1 + \frac{1}{2}z + \frac{1}{6}z^2 + \dots - z - z^3 - \dots \\
 &= \frac{1}{z} + 1 - \frac{1}{2}z - \frac{5}{6}z^2 - \dots
 \end{aligned}$$

(b) (i)  $z^2 > 2$

$$f(z) = \frac{1}{z^2+1} - \frac{1}{z^2+2} = \frac{1}{z^2} \frac{1}{1+z^2} - \frac{1}{z^2} \frac{1}{1+2z^2}$$

$$= \frac{1}{z^2} (1 - z^2 + z^4 - z^6 + \dots - 1 + 2z^2 - 4z^4 + 6z^6 - \dots)$$

$$= \frac{1}{z^2} (z^2 - 3z^4 + 5z^6 - \dots)$$

$$= z^{-4} - 3z^{-6} + \dots = \sum_{i=2}^{\infty} (-1)^i (2i-3) z^{-2i}$$

(ii)  $1 < z^2 < 2$

$$f(z) = \frac{1}{z^2+1} - \frac{1}{z^2+2} = \frac{1}{z^2} \frac{1}{1+z^2} - \frac{1}{2} \frac{1}{1+\frac{1}{2}z^2}$$

$$= \frac{1}{z^2} (1 - z^2 + z^4 - z^6 + \dots) - \frac{1}{2} (1 - \frac{1}{2}z^2 + \frac{1}{4}z^4 - \frac{1}{8}z^6 + \dots)$$

$$= z^{-2} - z^{-4} + z^{-6} + \dots - \frac{1}{2} + \frac{1}{4}z^2 - \frac{1}{8}z^4 + \dots = \sum_{i=0}^{\infty} (-1)^i [z^{-(2i+2)} - (\frac{1}{2})^{i+1} z^{2i}]$$

(iii)  $0 < z^2 < 1$

$$f(z) = \frac{1}{z^2+1} - \frac{1}{z^2+2} = \frac{1}{1+z^2} - \frac{1}{2} \frac{1}{1+\frac{1}{2}z^2}$$

$$= (1 - z^2 + z^4 - z^6 + \dots) - \frac{1}{2} (1 - \frac{1}{2}z^2 + \frac{1}{4}z^4 - \frac{1}{8}z^6 + \dots)$$

$$= \frac{1}{2} - \frac{3}{4}z^2 + \frac{7}{8}z^4 - \frac{15}{16}z^6 + \dots = \sum_{i=0}^{\infty} (-1)^i [1 - (\frac{1}{2})^{i+1}] z^{2i}$$

### 3.3 PR 3.3

(a) Expand the following functions in a Laurent series about  $z = 0$  ( $5!$  can be typed as factorial(5)):

- (i)  $\frac{\sin z}{z}$  (first three terms, ascending powers);
- (ii)  $\frac{\cosh(1/z)}{z}$  (first three terms, descending powers);
- (iii)  $\frac{e^{z^2}}{z^4}$  (first three terms, ascending powers).

(b) For each of the Laurent series in part (a), state the type of singularity, giving the order of any poles.

(a) (i)  $\frac{\sin z}{z} = \frac{1}{z} (z - \frac{1}{6}z^3 + \frac{1}{120}z^5 - \dots) = 1 - \frac{1}{6}z^2 + \frac{1}{120}z^4 - \dots$

(ii)  $\frac{\cosh(\frac{1}{z})}{z} = \frac{1}{2z} (e^{\frac{1}{z}} + e^{-\frac{1}{z}}) = \frac{1}{2z} (1 + \frac{1}{z} + \frac{1}{2!z^2} + \dots + 1 - \frac{1}{z} + \frac{1}{2!z^2} + \dots)$

$$= \frac{1}{z} (1 + \frac{1}{2!z^2} + \frac{1}{24!z^4} + \dots) = z^{-1} + \frac{1}{2}z^{-3} + \frac{1}{24}z^{-5} + \dots$$

$$\text{iii) } \frac{e^z}{z^4} = \frac{1}{z^4} \left( 1 + z + \frac{1}{2}z^2 + \dots \right)$$

$$= z^{-4} + z^{-3} + \frac{1}{2}z^{-2} + \dots$$

(b) (i) removable

(ii) essential

(iii) pole of order 4

### 3.4 PR 3.4

Find the residues of the following functions at the points indicated:

(a)  $\frac{1}{(e^z - 1)^2}$  at  $z = 2n\pi i$

(b)  $\frac{z^3}{z^2 + 1}$  at  $z = \pm i$



(c)  $e^z \tan z$  at  $z = 3\pi/2$

(a) let  $p = 1$ ,  $q = (e^z - 1)^2$ ,  $z_0 = 2n\pi i$

$$\therefore p' = 0$$

$$q' = 2e^z(e^z - 1)$$

$$q'' = 2e^z(e^z - 1) + 2e^z e^z = 2e^z(2e^z - 1)$$

$$q''' = 2e^z(2e^z - 1) + 2e^z(2e^z) = 2e^z(4e^z - 1)$$

$$\therefore q'(z_0) = 0$$

$$q''(z_0) = 2 \neq 0$$

$$q'''(z_0) = 6 \neq 0$$

$$\therefore \operatorname{Res}_{z=2n\pi i} \frac{1}{(e^z - 1)^2} = \frac{2p}{q''} \left( \frac{p'}{p} - \frac{q'''}{3q''} \right) = \frac{2}{2} \left( 0 - \frac{6}{6} \right) = -1$$

(b)  $p = z^2$ ,  $q = z^2 + 1$

$$\therefore q' = 2z$$

At  $z_0 = \pm i$ ,  $q(z_0) = 0$  and  $q'(z_0) = \pm 2i \neq 0$

$$\therefore \operatorname{Res}_{z=i} \frac{z^2}{z^2 + 1} = \frac{p(z_0)}{q'(z_0)} = \frac{-i}{2i} = -\frac{1}{2}$$

$$\operatorname{Res}_{z=-i} \frac{z^2}{z^2 + 1} = \frac{p(z_0)}{q'(z_0)} = \frac{i}{-2i} = -\frac{1}{2}$$

(c)  $p = e^z \sin z$ ,  $q = \cos z$

$$\therefore q' = -\sin z$$

$$\therefore p(z_0) = -e^{\frac{3}{2}\pi i}$$

$$q(z_0) = 0$$

$$q'(z_0) = 1$$

$$\therefore \operatorname{Res}_{z=\frac{3}{2}\pi i} e^z \tan z = \frac{-e^{\frac{3}{2}\pi i}}{1} = -e^{\frac{3}{2}\pi i}$$

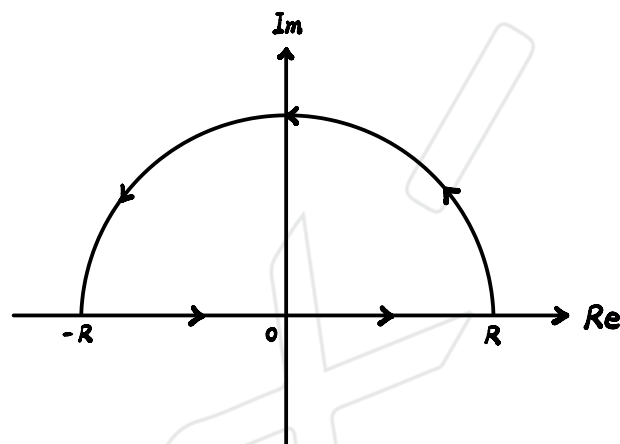
### 3.5 PR 3.5

By integrating the function

$$\frac{e^{iz}}{(z^2 + a^2)(z^2 + b^2)}$$

around a semicircular contour, show that

$$\int_{-\infty}^{\infty} \frac{\cos x}{(x^2 + a^2)(x^2 + b^2)} dx = \frac{\pi}{a^2 - b^2} \frac{a e^{-b} - b e^{-a}}{ab}.$$



$$f(z) = \frac{e^{iz}}{(z^2 + a^2)(z^2 + b^2)}$$

Therefore,

$$\oint_C f(z) = 2\pi i \left[ \text{Res}_{z=ia} f(z) + \text{Res}_{z=ib} f(z) \right]$$

where let  $p = e^{iz}$ ,  $q = (z^2 + a^2)(z^2 + b^2)$

$$\therefore q' = 2z(z^2 + b^2) + (z^2 + a^2)(2z) = 2z(2z^2 + a^2 + b^2)$$

$$\therefore p(ia) = e^{-a}$$

$$p(ib) = e^{-b}$$

$$q(ia) = q(ib) = 0$$

$$q'(ia) = 2ia(-2a^2 + a^2 + b^2) = 2ia(b^2 - a^2)$$

$$q'(ib) = 2ib(-2b^2 + a^2 + b^2) = 2ib(a^2 - b^2)$$

$$\therefore \operatorname{Res}_{z=ia} f(z) = \frac{e^{-a}}{2ia(b^2 - a^2)}$$

$$\operatorname{Res}_{z=ib} f(z) = \frac{e^{-b}}{2ib(a^2 - b^2)}$$

$$\begin{aligned} \therefore \oint_C f(z) &= 2\pi i \left[ \frac{e^{-a}}{2ia(b^2 - a^2)} + \frac{e^{-b}}{2ib(a^2 - b^2)} \right] \\ &= 2\pi i \frac{ibe^{-a} -iae^{-b}}{2ab(a^2 - b^2)} \\ &= \frac{\pi}{ab} \frac{ae^{-b} - be^{-a}}{a^2 - b^2} \end{aligned}$$

Meanwhile,

$$\oint_C f(z) = \int_{-R}^R \frac{e^{ix}}{(x^2 + a^2)(x^2 + b^2)} dx + \int_{\Gamma_R} \frac{e^{iz}}{(z^2 + a^2)(z^2 + b^2)} dz$$

for  $R \rightarrow \infty$ ,

$$\left| \int_{\Gamma_R} \frac{e^{iz}}{(z^2 + a^2)(z^2 + b^2)} dz \right| \leq \int_{\Gamma_R} \frac{|e^{iz}|}{|(z^2 + a^2)(z^2 + b^2)|} |dz| = \int_{\Gamma_R} \frac{1}{(R^2 + a^2)(R^2 + b^2)} \pi R \rightarrow 0$$

Therefore,

$$\oint_C f(z) = \int_{-\infty}^{\infty} \frac{e^{ix}}{(x^2 + a^2)(x^2 + b^2)} dx = \frac{\pi}{ab} \frac{ae^{-b} - be^{-a}}{a^2 - b^2}$$

### 3.6 PR 3.6

Evaluate by contour integration:

(a)  $\int_0^{2\pi} \frac{\cos \theta}{5 - 4 \cos \theta} d\theta;$

(b)  $\int_0^\infty \frac{dx}{1+x^n}$ , where  $n$  is an integer greater than 2;

(c)  $\int_{-\infty}^\infty \frac{\sin x}{x(x^2+1)} dx.$

(a) let  $z = e^{i\theta}$ , therefore  $\cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta}) = \frac{1}{2}(z + z^{-1})$

$\therefore \frac{dz}{d\theta} = ie^{i\theta} \quad \therefore \frac{d\theta}{dz} = \frac{1}{ie^{i\theta}} = -iz^{-1}$

$$\int_0^{2\pi} \frac{\cos \theta}{5 - 4 \cos \theta} d\theta = \oint_C \frac{z + z^{-1}}{10 - 4(z + z^{-1})} (-iz^{-1}) dz = -i \oint_C \frac{1 + z^{-2}}{10 - 4(z + z^{-1})} dz$$

$$= i \oint_C \frac{z^2 + 1}{4z^2 - 10z^2 + 4z} dz = i \oint_C \frac{z^2 + 1}{2z(2z - 1)(z - 2)} dz$$

$$= i \cdot 2\pi i \left[ \operatorname{Res}_{z=0} f(z) + \operatorname{Res}_{z=\frac{1}{2}} f(z) + \operatorname{Res}_{z=2} f(z) \right] dz$$

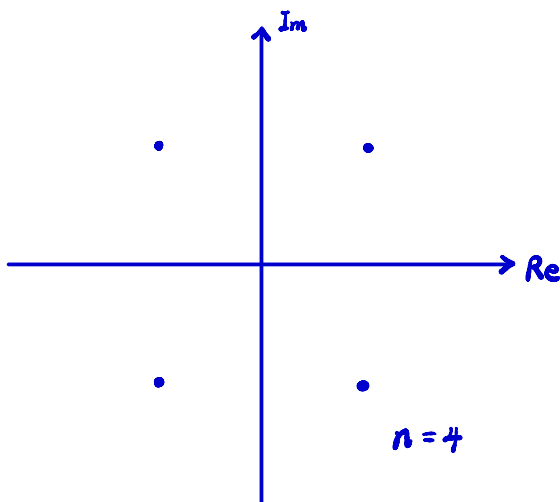
$$= -2\pi \left[ \frac{1}{4} + \frac{\frac{5}{4}}{1 \cdot 2 \cdot (-\frac{3}{2})} + 0 \right] \quad (\operatorname{Res}_{z=2} f(z) \text{ outside the contour})$$

$$= -2\pi \left( \frac{1}{4} - \frac{5}{12} \right) = \frac{1}{3}\pi$$

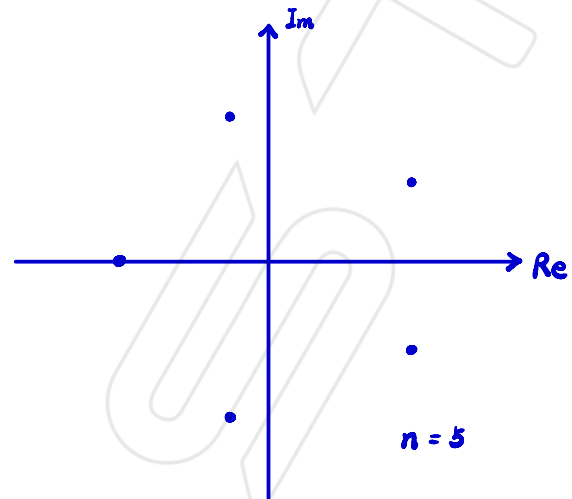
(b) Let  $f(z) = \oint_C \frac{1}{1+z^n} dz$

where poles at  $1 + z_0^n = 0 \quad \therefore z_0^n = e^{in\theta_0} = -1 \quad \therefore \theta_0 = \frac{\pi}{n} \quad \therefore z_0 = e^{i\frac{\pi}{n}} = e^{i(\frac{\pi + 2k\pi}{n})}$

for  $n$  even.

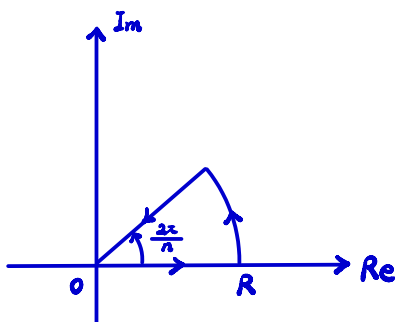


for  $n$  odd.



Therefore, to include only one isolated pole in contour (for the ease of calculation),

set contour as:



$$\begin{aligned}
 f(z) &= \oint_C \frac{1}{1+z^n} dz \\
 &= \int_0^R \frac{1}{1+x^n} dx + \int_{rR} \frac{1}{1+z^n} dz + \int_R^0 \frac{1}{1+(te^{i\frac{2\pi}{n}})^n} (e^{i\frac{2\pi}{n}}) dt \\
 &= \int_0^R \frac{1}{1+x^n} dx + \int_{rR} \frac{1}{1+z^n} dz + \int_R^0 \frac{e^{i\frac{2\pi}{n}}}{1+t^n} dt \\
 &= (1 - e^{i\frac{2\pi}{n}}) \int_0^R \frac{1}{1+x^n} dx + \int_{rR} \frac{1}{1+z^n} dz
 \end{aligned}$$

for  $R \rightarrow \infty$ .

$$\left| \int_{rR} \frac{1}{1+z^n} dz \right| \leq \int_{rR} \frac{1}{|z^n - (-1)|} |dz| \leq \int_{rR} \frac{1}{|z^n| - |(-1)|} |dz| = \int_{rR} \frac{\frac{2\pi}{n} R}{R^n - 1} \rightarrow 0$$

$$\therefore f(z) = \oint_C \frac{1}{1+z^n} dz = (1 - e^{i\frac{2\pi}{n}}) \int_0^R \frac{1}{1+x^n} dx$$

Meanwhile, at  $z_0 = e^{i\frac{\pi}{n}}$  inside the contour,

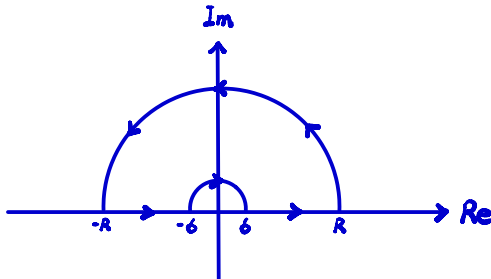
$$\text{let } p=1, q=1+z^n \quad \therefore q(z_0) = 0 \text{ but } q'(z_0) = nz^{n-1}|_{z=z_0} = ne^{i\frac{\pi}{n}(n-1)} = ne^{i\pi} e^{-i\frac{\pi}{n}} = -ne^{-i\frac{\pi}{n}}$$

$$\therefore f(z) = \oint_C \frac{1}{1+z^n} dz = 2\pi i \frac{1}{-ne^{-i\frac{\pi}{n}}} = -\frac{2\pi i e^{i\frac{\pi}{n}}}{n}$$

$$\therefore \int_0^{\infty} \frac{1}{1+x^n} dx = \frac{1}{1-e^{i\frac{2\pi}{n}}} \left( -\frac{2\pi i e^{i\frac{2\pi}{n}}}{n} \right) = \frac{-2\pi i}{n(e^{-i\frac{2\pi}{n}} - e^{i\frac{2\pi}{n}})} = \frac{-2\pi i}{-2ni \sin(\frac{2\pi}{n})} = \frac{\pi}{n \sin(\frac{2\pi}{n})}$$

(c) let  $f(z) = \oint \frac{e^{iz}}{z(z^2+1)} dz$

integrate  $f(z)$  around the contour shown below:



$\int_R^{\delta} \frac{e^{ix} e^{i\pi/2}}{x e^{i\pi/2} (x^2+1)} e^{i\pi/2} dx$  and let  $x' = -x$

$$f(z) = \oint \frac{e^{iz}}{z(z^2+1)} dz = \int_{\delta}^R \frac{e^{ix}}{x(x^2+1)} dx + \int_{-R}^{-\delta} \frac{e^{ix}}{x(x^2+1)} dx + \int_{\Gamma_R} \frac{e^{iz}}{z(z^2+1)} dz + \int_{\Gamma_{\delta}} \frac{e^{iz}}{z(z^2+1)} dz$$

$$\text{for } R \rightarrow \infty, \left| \int_{\Gamma_R} \frac{e^{iz}}{z(z^2+1)} dz \right| \leq \int_{\Gamma_R} \frac{|e^{iz}|}{|z||z^2+1|} |dz| \leq \int_{\Gamma_R} \frac{|e^{iz}|}{|z|(|z^2|-1)} |dz|$$

$$= \int_{\Gamma_R} \frac{\pi}{R^2-1} \rightarrow 0$$

$$\text{for } \delta \rightarrow 0, \int_{\Gamma_{\delta}} \frac{e^{iz}}{z(z^2+1)} dz = - \int_0^{\pi} \frac{e^{i\delta e^{i\theta}}}{\delta e^{i\theta} (\delta e^{i\theta} + 1)} i\delta e^{i\theta} d\theta = - \int_0^{\pi} i \frac{e^{i\delta e^{i\theta}}}{(\delta e^{i\theta} + 1)} d\theta \rightarrow -i \int_0^{\pi} d\theta = -i\pi$$

$$\therefore f(z) = \oint \frac{e^{iz}}{z(z^2+1)} dz = \int_{-\infty}^{\infty} \frac{e^{ix}}{x(x^2+1)} dx - i\pi = \int_{-\infty}^{\infty} \frac{\cos x}{x(x^2+1)} dx + i \int_{-\infty}^{\infty} \frac{\sin x}{x(x^2+1)} dx - i\pi$$

$$\therefore f(z) = \oint \frac{e^{iz}}{z(z^2+1)} dz = \int_{-\infty}^{\infty} \frac{\cos x}{x(x^2+1)} dx + i \left( \int_{-\infty}^{\infty} \frac{\sin x}{x(x^2+1)} dx - \pi \right)$$

Meanwhile,

$$\oint \frac{e^{iz}}{z(z^2+1)} dz = 2\pi i \left[ \frac{e^{iz}}{(z^2+1)+2z^2} \Big|_{z=i} \right]$$

$$= 2\pi i \left( -\frac{1}{2} e^{-1} \right)$$

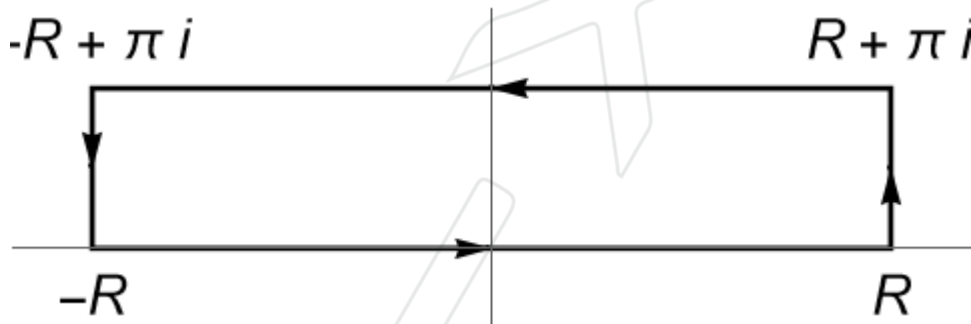
$$= i(-\pi e^{-1})$$

$$\therefore \int_{-\infty}^{\infty} \frac{\cos x}{x(x^2+1)} dx + i \left( \int_{-\infty}^{\infty} \frac{\sin x}{x(x^2+1)} dx - \pi \right) = i(-\pi e^{-1})$$

$$\therefore \int_{-\infty}^{\infty} \frac{\sin x}{x(x^2+1)} dx - \pi = -\pi e^{-1}$$

$$\therefore \int_{-\infty}^{\infty} \frac{\sin x}{x(x^2+1)} dx = \pi - \pi e^{-1}$$

3.7 PR 3.7



Using the contour shown, show that

$$\int_{-\infty}^{\infty} \frac{\cos x}{e^x + e^{-x}} dx = \frac{\pi}{e^{\pi/2} + e^{-\pi/2}}.$$

$$\text{Let } f(z) = \oint_C \frac{e^{iz}}{e^z + e^{-z}} dz$$

$$= \int_{-R}^R \frac{e^{ix}}{e^x + e^{-x}} dx + \int_0^{\pi i} \frac{e^{iR} e^{-y}}{e^R e^{iy} + e^{-R} e^{-iy}} dy$$

$$+ \int_R^{-R} \frac{e^{iR} e^{-x}}{e^R e^{ix} + e^{-R} e^{-ix}} dx + \int_{\pi i}^0 \frac{e^{-iR} e^{-y}}{e^{-R} e^{iy} + e^R e^{-iy}} dy$$

$$\text{for } R \rightarrow \infty, f(z) = \int_{-\infty}^{\infty} \frac{e^{ix}}{e^x + e^{-x}} dx - \int_{-\infty}^{\infty} \frac{e^{ix} e^{-x}}{e^R e^{ix} + e^{-R} e^{-ix}} dx$$

$$\text{where } \int_{-\infty}^{\infty} \frac{e^{ix} e^{-x}}{e^x e^{xi} + e^{-x} e^{-xi}} dx = \int_{-\infty}^{\infty} \frac{e^{ix} e^{-x}}{-(e^x + e^{-x})} dx \rightarrow -e^{-x} \int_{-\infty}^{\infty} \frac{e^{ix}}{e^x + e^{-x}} dx$$

$$\text{therefore, } f(z) = (1 + e^{-z}) \int_{-\infty}^{\infty} \frac{e^{ix}}{e^x + e^{-x}} dx$$

Meanwhile, for isolated pole  $z = \frac{\pi}{2}i$  in the contour

$$\text{let } p = e^{ix}, \quad q = e^x + e^{-x} \quad \therefore q' = e^x - e^{-x}$$

$$\therefore f(z) = 2\pi i \left[ \frac{e^{\frac{\pi}{2}i}}{e^{\frac{\pi}{2}i} - e^{-\frac{\pi}{2}i}} \right] = 2i \left( \frac{\pi e^{\frac{\pi}{2}i}}{2} \right) = i\pi e^{\frac{\pi}{2}i}$$

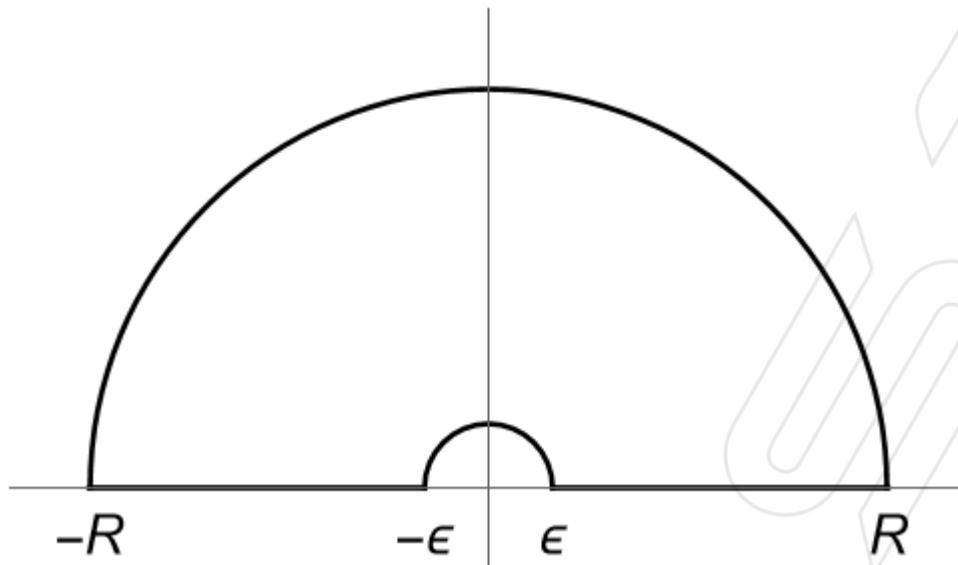
$$\therefore \int_{-\infty}^{\infty} \frac{e^{ix}}{e^x + e^{-x}} dx = \frac{\pi e^{-\frac{\pi}{2}i}}{1 + e^{-\pi}} = \frac{\pi}{e^{\frac{\pi}{2}} + e^{-\frac{\pi}{2}}}$$

$$\therefore \int_{-\infty}^{\infty} \left( \frac{\cos x}{e^x + e^{-x}} + i \frac{\sin x}{e^x + e^{-x}} \right) dx = \frac{\pi}{e^{\frac{\pi}{2}} + e^{-\frac{\pi}{2}}}$$

$$\therefore \int_{-\infty}^{\infty} \frac{\cos x}{e^x + e^{-x}} dx = \frac{\pi}{e^{\frac{\pi}{2}} + e^{-\frac{\pi}{2}}}$$



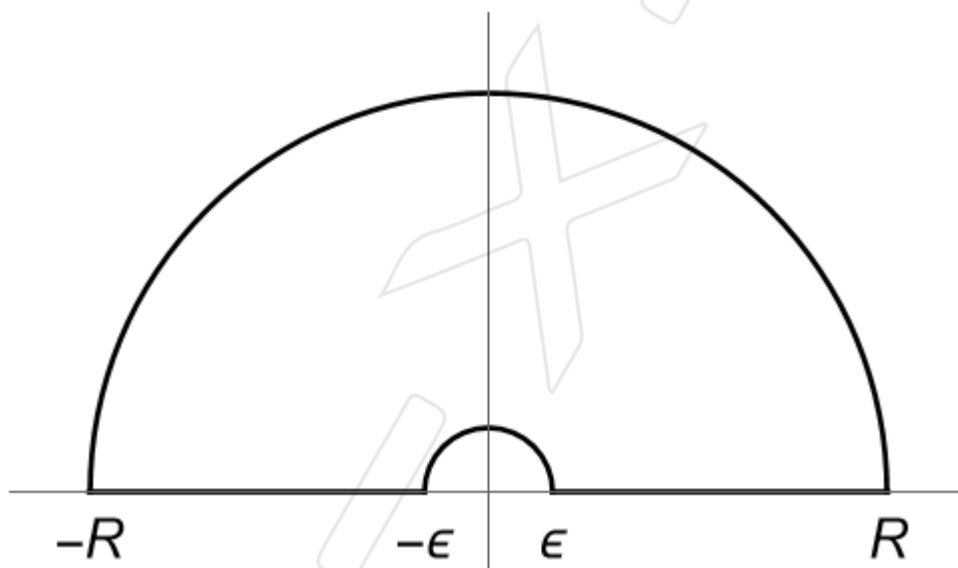
### 3.8 PR 3.8



(a)

Using the contour shown, show that

$$\int_0^{\infty} \frac{(\log x)^2}{1+x^2} dx = \frac{\pi^3}{8}.$$



(b)

Using the contour shown, show that

$$\int_0^{\infty} \frac{\log x}{(1+x^2)^2} dx = -\frac{\pi}{4}.$$

(a) let  $f(z) = \oint_{\epsilon} \frac{(\ln z)^2}{1+z^2} dz$

$$= \int_{\epsilon}^R \frac{(\ln x)^2}{1+x^2} dx + \oint_{R\epsilon} \frac{(\ln z)^2}{1+z^2} dz + \int_R^{\epsilon} \frac{(\ln x + i\pi)^2}{1+x^2} e^{i\pi} dx - \oint_{\epsilon} \frac{(\ln z)^2}{1+z^2} dz$$

for  $R \rightarrow \infty$ ,

$$\left| \oint_{\Gamma_R} \frac{(\ln z)^2}{1+z^2} dz \right| \leq \oint_{\Gamma_R} \frac{|(\ln z)^2|}{|z^2 - (-1)|} |dz| \leq \oint_{\Gamma_R} \frac{|(\ln R + i\theta)^2|}{|z^2 - 1|} |dz| \leq \oint_{\Gamma_R} \frac{(\ln R)^2 + \pi^2}{R^2 - 1} \pi R \rightarrow 0$$

for  $\varepsilon \rightarrow 0$

$$\left| \oint_{\delta\varepsilon} \frac{(\ln z)^2}{1+z^2} dz \right| \leq \oint_{\delta\varepsilon} \frac{|(\ln z)^2|}{|1 - (-z^2)|} |dz| \leq \oint_{\delta\varepsilon} \frac{|(\ln \varepsilon + i\theta)^2|}{1 - |z^2|} |dz| \leq \oint_{\delta\varepsilon} \frac{(\ln \varepsilon)^2 + \pi^2}{\varepsilon^2 - 1} \pi \varepsilon \rightarrow 0$$

$$\therefore f(z) = 2 \int_{-\infty}^{\infty} \frac{(\ln x)^2}{1+x^2} dx - \int_{-\infty}^{\infty} \frac{\pi^2}{1+x^2} dx + i \left( \int_{-\infty}^{\infty} \frac{2\pi \ln x}{1+x^2} dx \right)$$

Meanwhile, at isolated singular point  $z = i$

$$f(z) = \oint_C \frac{(\ln z)^2}{1+z^2} dz$$

$$= 2\pi i \left[ \frac{(\ln i)^2}{2i} \right]$$

$$= 2\pi i \left[ \frac{(\ln i + \frac{\pi}{2}i)^2}{2i} \right]$$

$$= \pi \left( \frac{\pi}{2}i \right)^2 = -\frac{1}{4}\pi^2$$

$$\therefore 2 \int_{-\infty}^{\infty} \frac{(\ln x)^2}{1+x^2} dx - \int_{-\infty}^{\infty} \frac{\pi^2}{1+x^2} dx + i \left( \int_{-\infty}^{\infty} \frac{2\pi \ln x}{1+x^2} dx \right) = -\frac{1}{4}\pi^2$$

$$\therefore 2 \int_{-\infty}^{\infty} \frac{(\ln x)^2}{1+x^2} dx - \int_{-\infty}^{\infty} \frac{\pi^2}{1+x^2} dx = 2 \int_{-\infty}^{\infty} \frac{(\ln x)^2}{1+x^2} dx - \frac{\pi^2}{2} = -\frac{1}{4}\pi^2$$

$$\therefore \int_{-\infty}^{\infty} \frac{(\ln x)^2}{1+x^2} dx = \frac{1}{8}\pi^2$$

b) let  $f(z) = \oint_z \frac{\ln z}{(1+z^2)^2} dz$

$$= \int_{\varepsilon}^R \frac{\ln x}{(1+x^2)^2} dx + \oint_{\Gamma_R} \frac{\ln z}{(1+z^2)^2} dz$$

$$+ \int_R^{\varepsilon} \frac{\ln x + i\pi}{(1+x^2)^2} e^{i\pi} dx + \oint_{\delta\varepsilon} \frac{\ln z}{(1+z^2)^2} dz$$

$$= 2 \int_{\varepsilon}^R \frac{\ln x}{(1+x^2)^2} dx + i \int_{\varepsilon}^R \frac{\pi}{(1+x^2)^2} dx$$

$$+ \oint_{\Gamma_R} \frac{\ln z}{(1+z^2)^2} dz + \oint_{\delta\varepsilon} \frac{\ln z}{(1+z^2)^2} dz$$

for  $R \rightarrow \infty$ ,

$$\left| \oint_{\Gamma_R} \frac{\ln z}{(1+z^2)^2} dz \right| \leq \oint_{\Gamma_R} \frac{|\ln z|}{|1+z^2|^2} |dz| \leq \oint_{\Gamma_R} \frac{\sqrt{(\ln R)^2 + \pi^2}}{(R^2-1)^2} \pi R \rightarrow 0$$

for  $\epsilon \rightarrow 0$ ,

$$\left| \oint_{\Gamma_\epsilon} \frac{\ln z}{(1+z^2)^2} dz \right| \leq \oint_{\Gamma_\epsilon} \frac{|\ln z|}{|1+z^2|^2} |dz| \leq \oint_{\Gamma_\epsilon} \frac{\sqrt{(\ln \epsilon)^2 + \pi^2}}{(1-\epsilon^2)^2} \pi \epsilon \rightarrow 0$$

$$\therefore f(z) = 2 \int_0^\infty \frac{\ln x}{(1+x^2)^2} dx + i \int_0^\infty \frac{\pi}{(1+x^2)^2} dx$$

Meanwhile, within the contour, there is an isolated pole  $z_0 = i$

$$\text{Let } p = \ln z, \quad q = (1+z^2)^2 \quad \therefore p(z_0) = \ln i = \ln 1 + i \frac{\pi}{2} = \frac{\pi}{2}i, \quad q(z_0) = 0$$

$$\therefore p' = \frac{1}{z}$$

$$\therefore p'(z_0) = \frac{1}{i} = -i$$

$$q' = 4z(1+z^2) = 4z^3 + 4z$$

$$q'(z_0) = 0$$

$$q'' = 12z^2 + 4$$

$$q''(z_0) = -8$$

$$q''' = 24z$$

$$q'''(z_0) = 24i$$

$$\therefore f(z) = 2\pi i \cdot \frac{\pi i}{-8} \left( -\frac{2}{\pi} - \frac{24i}{-24} \right) = \frac{1}{4}\pi^2 \left( -\frac{2}{\pi} + i \right) = -\frac{1}{2}\pi + \frac{1}{4}\pi^2 i$$

$$\therefore 2 \int_0^\infty \frac{\ln x}{(1+x^2)^2} dx + i \int_0^\infty \frac{\pi}{(1+x^2)^2} dx = -\frac{1}{2}\pi + \frac{1}{4}\pi^2 i$$

$$\therefore \int_0^\infty \frac{\ln x}{(1+x^2)^2} dx = -\frac{1}{4}\pi$$

### 3.9 PR 3.9

By amending the derivation of the Blasius law for the hydrodynamic force, show that the *moment* for the complex potential  $\Omega(z)$  is

$$\begin{aligned} M &= -\frac{\rho}{2} \oint |\Omega'(z)|^2 (x dx + y dy) \\ &= \text{Re} \left( -\frac{\rho}{2} \oint |\Omega'(z)|^2 z d\bar{z} \right) \\ &= \text{Re} \left( -\frac{\rho}{2} \oint \Omega'(z)^2 z dz \right). \end{aligned}$$

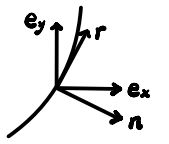
$$\Omega(x+iy) = \phi(x+iy) + i\psi(x+iy)$$

$$\therefore V = \frac{d\Omega}{dz} = u - iv$$

$$\therefore \bar{V} = u + iv$$

$$\therefore V(z) = \overline{U_0} + C_1 z^{-1} + C_2 z^{-2} + \dots$$

for unit vectors.



$$\begin{aligned}\vec{r} &= \frac{dx}{ds} \vec{e}_x + \frac{dy}{ds} \vec{e}_y \\ \vec{n} &= \frac{dy}{ds} \vec{e}_x - \frac{dx}{ds} \vec{e}_y\end{aligned}$$

According to 2-D Bernoulli equation,  $P = p_0 + \frac{1}{2}\rho(u^2 + v^2) = p_0 + \frac{1}{2}\rho|\Omega'(z)|^2$

$$\therefore F = \oint_C P n ds = -\oint_C P dy \vec{e}_x + \oint_C P dx \vec{e}_y$$

$$\therefore F_x = -\oint_C P dy$$

$$F_y = \oint_C P dx$$

$$\therefore M = F_x \cdot y + F_y \cdot x$$

$$= \oint_C P (x dx + y dy)$$

$$= p_0 \oint_C (x dx + y dy) + \frac{1}{2}\rho \oint_C |\Omega'(z)|^2 (x dx + y dy)$$

$$= \frac{1}{2}\rho \oint_C |\Omega'(z)|^2 (x dx + y dy)$$

This is because

$$\oint_C x dx = \int_{t_0}^{t_1} x \frac{dx}{dt} dt = \int_{t_0}^{t_1} x \dot{x} dt = \left[ \frac{1}{2} x^2 \right]_{t_0}^{t_1} = \frac{1}{2} \{ x(t_1)^2 - x(t_0)^2 \} = 0$$

where  $x(t_1) = x(t_0)$  for closed-curve

same for  $\oint_C y dy$

$$\text{so } \oint_C x dx + y dy = 0$$

$$\therefore \oint_{\gamma} P z d\bar{z} = \oint_{\gamma} P(x+iy) d(x-iy) = \oint_{\gamma} P(xdx+ydy) + i \oint_{\gamma} P(ydx-xdy)$$

$$\therefore M = \operatorname{Re} \left( -\frac{1}{2} \rho \oint_{\gamma} |f(z)|^2 z d\bar{z} \right)$$

$$\therefore M = \operatorname{Re} \left( -\frac{1}{2} \rho \oint_{\gamma} \frac{dn}{dz} \frac{d\bar{n}}{dz} z d\bar{z} \right)$$

$$= \operatorname{Re} \left( -\frac{1}{2} \rho \oint_{\gamma} \frac{dn}{dz} d\bar{n} z \right)$$

along  $\gamma$ ,  $dy = 0$

$$\therefore d\bar{n} = d\phi - i dy = d\phi + i dy = dn$$

$$\therefore M = \operatorname{Re} \left( -\frac{1}{2} \rho \oint_{\gamma} \frac{dn}{dz} dn z \right) = \operatorname{Re} \left( -\frac{1}{2} \rho \oint_{\gamma} \left( \frac{dn}{dz} \right)^2 z dz \right)$$